



Keras

Keras

vs

FastAI



fastai



| Luiz Paludo



| ITAI 2376



| Houston Community College



| August 2026

INTRODUCTION

Keras and **FastAI** are Python libraries used to build and train deep learning models. They **simplify complex tasks** and help users **create accurate models** efficiently.

K Keras



Focuses on **simplicity, flexibility,** and **multi-framework support.**



Works with **multiple backends:** TensorFlow, PyTorch, and JAX.



Designed for **scalability** and production environments.



Good for both **beginners** and **experienced developers.**

VS

fastai



Focuses on **fast** and **accurate** models using modern best practices.



Built on PyTorch with a **high-level, layered API.**



Requires **less code** for common deep learning tasks.



Excellent for **learning, rapid prototyping,** and **research.**



Main Question:

Which library is more suitable for **different types** of deep learning users and projects?

BACKGROUND

A look at the origin, purpose, and ecosystem behind **Keras** and **FastAI**.

K Keras



Origin and Development

- Created in **2015** by **François Chollet**, a Google engineer.
- Designed as a high-level API to make deep learning more accessible.



Purpose

- Provide a simple, consistent, and user-friendly interface for building deep learning models.
- Focus on flexibility and ease of use across different platforms.



Ecosystem

- Supports multiple backends: TensorFlow, PyTorch, and JAX.
- Part of a larger ecosystem that includes TensorFlow and the wider ML community.

VS

fastai



Origin and Development

- Created in **2016** by **Jeremy Howard** and the **FastAI team**.
- Built on top of PyTorch to simplify deep learning workflows.



Purpose

- Make state-of-the-art deep learning techniques accessible with minimal code.
- Focus on rapid experimentation and achieving high accuracy quickly.



Ecosystem

- Built on PyTorch.
- Includes a high-level library (FastAI) that sits on top of PyTorch for faster development and best practices.

KEY FEATURES

A side-by-side look at the most important features of **Keras** and **FastAI**.

 Keras	 fastai
 Ease of Use Simple and intuitive API. Easy to learn for beginners.	 Ease of Use Very easy and fast to build models. High-level API with less code.
 Flexibility Highly flexible and customizable. Gives full control to advanced users.	 Built-in Best Practices Comes with default settings that follow modern best practices.
 Multi-Framework Support Keras 3 supports multiple backends: TensorFlow, PyTorch, and JAX.	 PyTorch Ecosystem Built on top of PyTorch. Takes advantage of its flexibility and tools.
 Built for Production Designed with deployment and scalability in mind.	 Fast Development Speeds up the deep learning workflow. Ideal for research and prototyping.
 Customization Allows detailed customization of models and training loops.	 Less Code, More Results Achieve a lot with less code. Focus more on solving problems.

VS



Key Takeaway:

Keras offers flexibility and control, while **FastAI** focuses on speed and productivity. Both are powerful but serve different needs.

APPLICATIONS & COMPARISON

Real-world use, usability, performance, support, and scalability compared side by side.

K Keras



Real-world Applications

- Used in production by companies like Google, Netflix, and NASA.
- Widely adopted in industry for reliable and scalable solutions.



Usability

- Beginner-friendly with a consistent and easy-to-understand API.
- Great documentation and community support.



Performance

- High performance with optimized backends.
- Users can choose the best backend for their needs.



Support & Community

- Large global community.
- Long-term support from the TensorFlow team and broad industry adoption.



Scalability

- Built for production and scales well from research to large systems.
- Suitable for enterprise-level applications.

VS

fastai



Real-world Applications

- Used in many research projects and startups.
- Popular for computer vision, NLP, and time series projects.



Usability

- Extremely easy to use and learn.
- Designed to get good results with minimal code.



Performance

- Excellent performance for research and quick experiments.
- Built-in functions help achieve high accuracy fast.



Support & Community

- Active and friendly community.
- Strong support from the FastAI team and open-source contributors.



Scalability

- Perfect for small to medium projects.
- Can be scaled, but more suitable for research and prototyping than large deployments.



Overall Comparison:

Keras is better for large-scale, production-ready systems.

FastAI is better for fast development, research, and achieving results quickly.

CODE COMPARISON & CONCLUSION

A simple example: build a basic model for image classification.

K Keras

Build and train a simple image classification model

```
1 from tensorflow import keras
2 from tensorflow.keras import layers
3
4 model = keras.Sequential([
5     layers.Conv2D(32, (3, 3), activation='relu',
6                 input_shape=(64, 64, 3)),
7     layers.MaxPooling2D((2, 2)),
8     layers.Flatten(),
9     layers.Dense(64, activation='relu'),
10    layers.Dense(10, activation='softmax')
11 ])
12
13 model.compile(optimizer='adam',
14             loss='sparse_categorical_crossentropy',
15             metrics=['accuracy'])
16 model.fit(train_ds, epochs=10, validation_data=val_ds)
```



More lines of code to build and train the model.
More **control and flexibility**.

fastai

Build and train a simple image classification model

```
1 from fastai.vision.all import *
2
3 dls = ImageDataLoaders.from_folder('data',
4                                   valid='valid', bs=32, size=64)
5
6 learn = cnn_learner(dls, resnet18, metrics=accuracy)
7
8 learn.fine_tune(10)
```



Much less code to build and train the model.
High-level API makes it **fast and easy**.

VS



FINAL QUESTION
Which one should
you choose?



Choose Keras if you want:

- Flexibility and control over every part.
- Multi-framework support (TensorFlow, PyTorch, JAX).
- Ideal for production and large-scale systems.



Choose FastAI if you want:

- Fast development with less code.
- Built-in best practices and powerful defaults.
- Great for research, prototyping, and achieving results quickly.



Bottom Line: Both Keras and FastAI are excellent. Your choice depends on your goals, experience, and the type of project you are working on.